

DVP&R; — VBF-T3R1-DVPR-001

Design Verification Plan & Report (virtual tests) — VBF-T3R1-DVPR-001

All tests executed with the bda simulation library, DFN mode, Chen2020 base parameters + design overrides (cell/p_v12.json). Verdicts are mechanical (bda log-evaluate, round 4).

#	Test item	Requirement	Method / protocol	Result	Verdict	Evidence
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1	Nominal capacity	≥ 2.0 Ah	1C_discharge, DFN	3.075 Ah	PASS	cell/r4_v12_1c.json:capacity_ah
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2	5C capacity retention	≥ 0.95	5C_discharge DFN / 1C_discharge DFN	0.97532	PASS	cell/r4_v12_derived.json:retention_5c
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3	Fast-charge plating	False (no plating)	4C_charge_45C, DFN, --plating	anode min +12.1 mV (>0)	PASS	cell/r4_v12_4c.json:anode_potential_v
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4	Max temperature	≤ 333.15 K	4C_charge_45C, DFN, --thermal lumped	329.77 K (56.62 C)	PASS	cell/r4_v12_derived.json:T_max_K
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5	Power density	≥ 4000.0 W/kg	calc-energy contract formula	32630 W/kg	PASS	cell/r4_v12_energy.json:power_density_w_kg
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Iteration history (log.jsonl)

Round	Candidates	Outcome
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1	Baseline (Chen2020 defaults)	FAIL: retention 0.0874, T_max 354.29 K, plated — power density/capacity OK. Failures are transport/architecture-scale -> Stage-3 fallback.
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2	V1-V6 single levers + combo	V6_ComboCeiling first full pass (retention 0.9782, T 324.91 K, no plating); Q_nom understated C-rates by ~15% (lesson recorded).
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3	V7-V11 refinements	Attribution: thicker anode insufficient (V7 retention fail), porosity boost load-bearing (V10 plated), moderate electrolyte erodes plating margin (V9 +0.6 mV), h-trade mapped (V8/V11).
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4	V12_BalancedCooling (h=20)	PASS 5/5 with balanced margins (3.38 K thermal, +12.1 mV plating) — final selection.
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Notes

- Stage-5 true DFT/MD endorsement skipped: real_compute=false (entry-0 meta); endorse entry records the skip.

- Aging/cycle-life not in task criteria; not tested (aging protocol available for follow-up).